Russell Wolfe

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Summary

Undergraduate researcher with hands-on Python development, machine learning, and high-performance computing experience applied to large scientific datasets. B.S. Mathematics candidate with strong statistical foundations and demonstrated ability to collaborate with senior researchers and communicate technical findings clearly.

Education

Colorado State University Expected May 2027
Bachelor of Science in Mathematics / Bachelor of Music in Violin Performance
Minor: Chemistry | 3.905 GPA
National Merit Scholar | U.S. Presidential Scholar Semifinalist

Experience

Undergraduate Student Researcher, Paton Research Group, Colorado State University 2026–Present

- Apply machine learning and computational methods to investigate chemical reactivity, processing large quantum mechanical datasets for analysis and model development
- Develop Python tools for data analysis, workflow automation, and visualization of simulation outputs
- Execute high-performance computing (HPC) workflows to run quantum chemical simulations, manage data pipelines, and analyze outputs
- Analyze machine-learned interatomic potentials trained on quantum mechanical reference data, evaluating model accuracy and refining inputs
- Collaborate with graduate researchers and postdoctoral scholars to advance predictive models; present findings clearly to senior team members

TtGG Course, Team Leader & Lab Assistant 2022

- Led a team of students through a six-week laboratory curriculum; directed preparation and sample organization
- Organized gel electrophoresis, PCR, and cell culture procedures, ensuring accuracy and reproducibility of results

Freelance Violinist 2021–Present

- Manage independent scheduling, client communication, and logistics for solo and ensemble engagements

Projects

TRIP50 Benchmarking: DFT vs. machine-learned potentials 2026

- Benchmarked DFT, machine-learned interatomic potentials, and a semiempirical method on 50 triplet-state reactions against coupled-cluster references; code and results are public

Skills

- **Programming:** Python, R, C++, LaTeX
- **Data Analysis:** Machine learning, statistical modeling, data cleaning and preparation, workflow automation
- **Visualization:** matplotlib, Microsoft Excel
- **Computing:** High-performance computing (HPC), Linux/command-line environments, Git
- **Scientific Software:** ORCA, Gaussian, g-xTB, ASE, RDKit
- **Tools:** Microsoft Office Suite, Google Suite
- **AI-assisted Workflows:** Prompt engineering, LLM tool use, agentic pipelines
- **Laboratory Techniques:** NMR, IR, titration, PCR, column chromatography, gel electrophoresis
- **Languages:** Seal of Biliteracy in Mandarin Chinese

Relevant Coursework

Biostatistics | Statistics: Theory and Practice | Linear Algebra | Calculus I–III | Ordinary Differential Equations | Real Analysis | Discrete Mathematics | General & Organic Chemistry I–II | Biochemistry | Physics I & II | Multivariate Analysis

Awards & Activities

- Dean's List, CSU College of Liberal Arts, all semesters
- Eagle Scout, Scouting USA (2023)
- Concertmaster, CSU Symphony (2024–Present)
- Winner, CSU Concerto Competition | 1st Place, 2022 Beethoven Club Young Artists Competition
- 16 ultramarathons (50–100 km); 1st overall 2022 Big Buffalo 50k, 3rd 2025 Colorado Marathon, 2nd 2026 Publix Atlanta Marathon